

[Aim: 100/100 in Maths]

आरंभ

TRIGONOMETRY

Lecture #4

#LP: $\sin(2A+B) = \sqrt{3}/2$ and $\tan(A+2B) = 1$, find the value of $\cot(4A-7B)$. [CBSE 2025]

A. $\sqrt{3}$ 29.48.1.

B. $1/\sqrt{3}$ 48.1.

C. $\sqrt{2}$ 16.1.

D. 2 5.9.1.

$$\sin(2A+B) = \frac{\sqrt{3}}{2}$$

$$\sin(2A+B) = \sin 60$$

on comparing

$$2A+B=60^\circ \quad \text{--- (I)}$$

$$\tan(A+2B) = 1$$

$$\tan(A+2B) = \tan 45$$

on comparing

$$A+2B=45 \quad \text{--- (II)}$$

$\times 2 \Rightarrow$

$$2A+4B=90$$

$$\begin{array}{r} 2A+B=60 \\ 2A+4B=90 \\ \hline -3B=-30 \end{array}$$

$$B=10$$

$$\begin{array}{r} 2A+10=60 \\ 2A=50 \end{array}$$

$$A=25^\circ$$

To find

$$\cot(4A-7B)$$

$$\cot[4(25)-7(10)]$$

$$\cot(100-70)$$

$$\cot 30^\circ$$

$$\Rightarrow \sqrt{3}$$

#LP: If $\tan(A+B) = \sqrt{3}$ and $\tan(A-B) = 1/\sqrt{3}$, find the value of $\tan A \sin(A+B) + \cos A \tan(A-B)$.

$$\tan(A+B) = \sqrt{3}$$

$$\tan(A+B) = \tan 60^\circ$$

on comparing, $\boxed{A+B=60^\circ}$ (I)

$$\tan(A-B) = \frac{1}{\sqrt{3}}$$

$$\tan(A-B) = \tan 30^\circ$$

$$\boxed{A-B=30^\circ}$$
 (II)

$$A = 45^\circ$$

$$B = 15^\circ$$

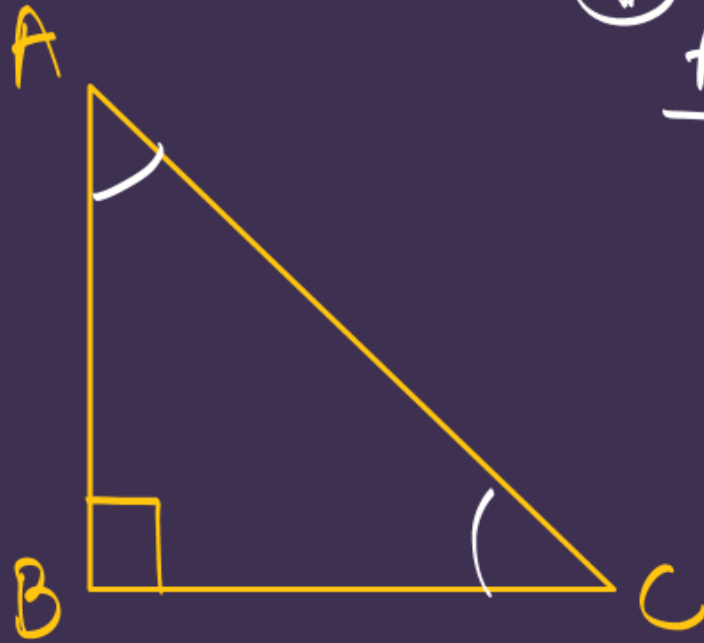
To find: $\tan A \sin(A+B) + \cos A \tan(A-B)$

$$\Rightarrow \tan 45^\circ \sin 60^\circ + \cos 45^\circ \tan 30^\circ$$

$$\Rightarrow (1) \left(\frac{\sqrt{3}}{2} \right) + \left(\frac{1}{\sqrt{2}} \right) \left(\frac{1}{\sqrt{3}} \right)$$

$$\Rightarrow \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}\sqrt{3}}$$

#LP: If ABC is a right angled triangle, right angled at B, then what is the value of $\sin(A+C)$?



④ Angle sum Prop of Δ

$$\angle A + \angle B + \angle C = 180^\circ$$



$$\angle A + 90 + \angle C = 180$$

$$A + C = 180 - 90$$

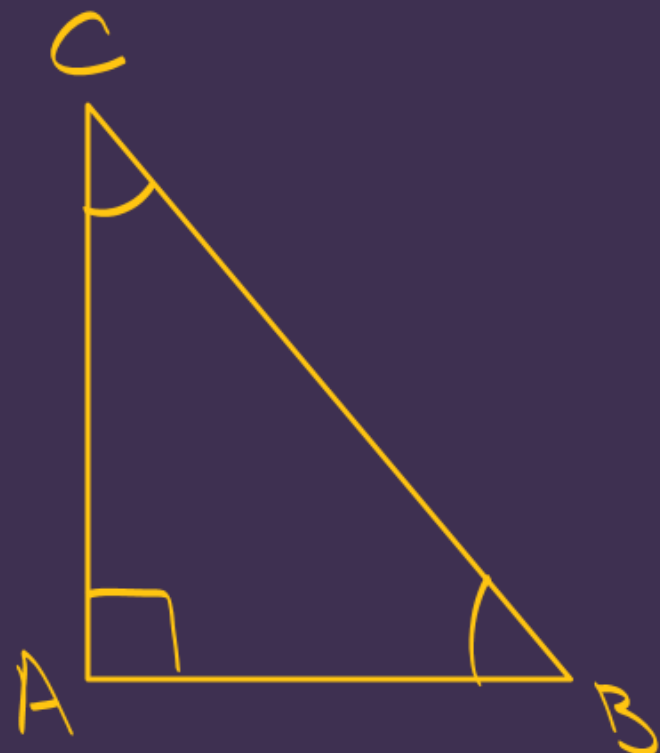
$$\boxed{A + C = 90^\circ}$$

To find:- $\sin(A+C)$

⇒ $\sin 90^\circ$
⇒ $\textcircled{1}$

#LP: In triangle ABC, if angle A is 90 degrees, find the value of $\tan \frac{B+C}{2}$

2



$$\text{Asp} \Rightarrow A + B + C = 180$$

$$90 + B + C = 180$$

$$B + C = 180 - 90$$

$$\boxed{B + C = 90^\circ} \checkmark$$

To find: $\tan \left(\frac{B+C}{2} \right)$

$$\tan \left(\frac{90}{2} \right) = \tan 45^\circ$$
$$= \boxed{1} \checkmark$$

#LP: In an acute angled triangle ABC, if $\tan(A+B-C)=1$ and $\sec(B+C-A)=2$, find the value of A, B and C.

$$\tan(A+B-C)=1$$

$$\tan(A+B-C)=\tan 45$$

$$A+B-C=45 \quad \text{--- (I)}$$

$$\sec(B+C-A)=2$$

$$\sec(B+C-A)=\sec 60^\circ$$

$$B+C-A=60^\circ \quad \text{--- (II)}$$

ASP of Δ

$$A+B+C=180 \quad \text{--- (III)}$$

$$\text{(I)} \rightarrow A+B-C=45$$

$$\text{(II)} \rightarrow -A+B+C=60$$

⊕

$$2B=105^\circ$$

$$B=\frac{105}{2}$$

$$\text{(I)} \rightarrow -A+B+C=60$$

$$\text{(III)} \rightarrow A+B+C=180$$

$$-1 \times A = +120$$

$$A=60^\circ \quad \checkmark$$

$$60 + \frac{105}{2} + C = 180$$

$$\frac{105}{2} + C = 180 - 60$$

$$\frac{105}{2} + C = 120$$

$$C = 120 - \frac{105}{2}$$

$$C = \frac{240 - 105}{2}$$

$$C = \frac{135}{2} \quad \checkmark$$

#LP: Express the trigonometry ratios **Sin A**, **Sec A** and **Tan A** in terms of **Cot A**. ✓

दिए गये terms से लिखना है $\rightarrow x$

$$\boxed{\cot A = x}$$

$$\frac{P}{B} = \frac{x}{1}$$

$$\boxed{B = xk}$$

$$\boxed{P = 1 \cdot k}$$

By Pyth

$$H^2 = P^2 + B^2$$

$$H^2 = (1k)^2 + (xk)^2$$

$$H^2 = k^2 + x^2 k^2$$

$$H^2 = k^2(1+x^2)$$

$$H = \frac{k^2(1+x^2)}{\sqrt{k^2(1+x^2)}} \Rightarrow$$

$$\boxed{H = k\sqrt{1+x^2}}$$

$$\sin A = \frac{P}{H} = \frac{k}{k\sqrt{1+x^2}} = \frac{1}{\sqrt{1+\cot^2 A}}$$

$$\sec A = \frac{H}{B} = \frac{k\sqrt{1+x^2}}{xk}$$

$$\boxed{\sec A = \frac{\sqrt{1+\cot^2 A}}{\cot A}}$$

$$\tan A = \frac{P}{B} = \frac{k}{xk} = \frac{1}{\cot A}$$

#LP: Express the ratios cosA, tanA and sec A in terms of sin A.

$$\text{let } \sin A = x$$

$$\frac{P}{H} = \frac{x}{1}$$

$$\text{let } P = xk \quad H = 1k$$

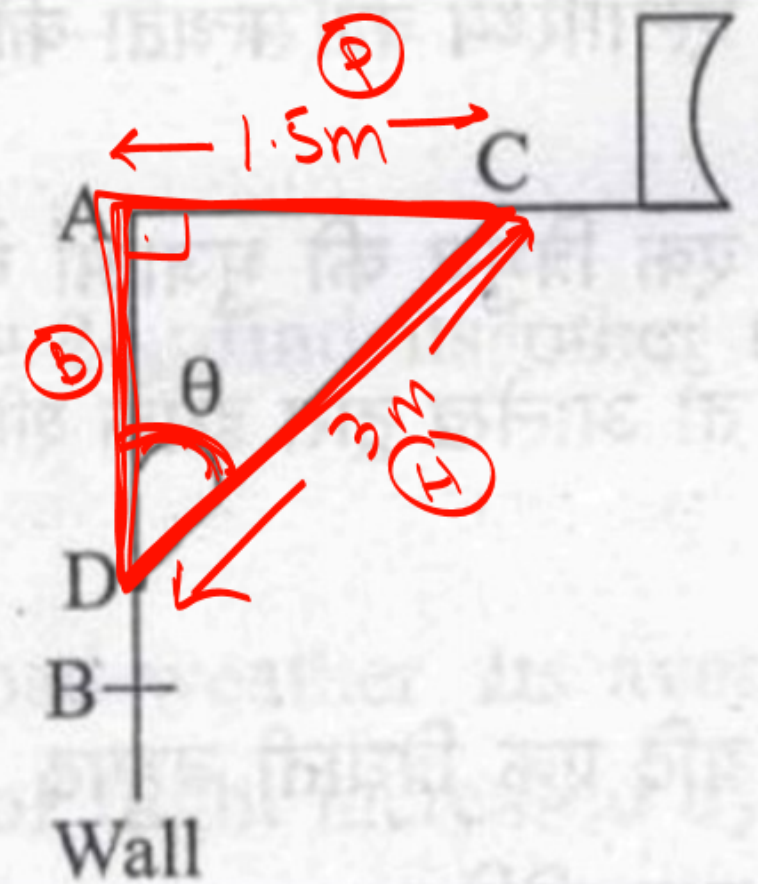
Pyth's

$$B = \sqrt{\quad}$$

#LP: CBSE 2020

23. The rod AC of a TV disc antenna is fixed at right angles to the wall AB and a rod CD is supporting the disc as shown in Fig. 4. If AC = 1.5 m long and CD = 3 m, find (i) $\tan \theta$ (ii) $\sec \theta + \operatorname{cosec} \theta$

Fig. 4



i) $\tan \theta = \frac{P}{B}$

Pytha $\rightarrow$ B ✓

#LP:

If θ is an acute angle and $\tan \theta + \cot \theta = 2$, then the value of $\sin^3 \theta + \cos^3 \theta$ is

A. 1

B. 2

C. $1/\sqrt{2}$ ✓

D. $\sqrt{2}$

$$1 + 1 = 2$$

$$\tan \theta = 1$$

$$\cot \theta = 1$$

$$\theta = 45^\circ$$

$$\theta = 45^\circ$$

To find

$$\sin^3 \theta + \cos^3 \theta$$

$$(\sin 45) + (\cos 45)$$

$$\Rightarrow \left(\frac{1}{\sqrt{2}}\right)^3 + \left(\frac{1}{\sqrt{2}}\right)^3$$

$$\Rightarrow 2 \left(\frac{1}{\sqrt{2}}\right)^3$$

$$\Rightarrow 2 \times \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}}$$

$$\Rightarrow \cancel{2} \times \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} \Rightarrow \left(\frac{1}{\sqrt{2}}\right)$$

Q: If $\tan \theta - \cot \theta = 0$, find the value of $\tan^{92} \theta + \cot^{69} \theta$?

(A) 1

(B) 2

(C) 4

(D) 0

$$\downarrow \quad \downarrow \\ 1 - 1 = 0$$

$$\tan \theta = 1 \quad \cot \theta = 1 \\ \boxed{\theta = 45} \quad \boxed{\theta = 45}$$

$$(\tan 45)^{92} + (\cot 45)^{69}$$

$$1^{92} + 1^{69}$$

$$\Rightarrow 1 + 1 \Rightarrow \textcircled{2}$$

#Q: If $\sin A - \cos A = 0$, find the value of $\frac{\cos^2 A + \sin^2 A}{\tan A}$

(A) 2

(B) $\frac{1}{2}$

~~(C) 1~~

(D) -1

$\sin A = 1$
 $A = 90^\circ$

$\cos A = 1$
 $A = 0^\circ$

$\sin A - \cos A = 0$

$$\sin A = \cos A$$

$$\frac{\sin A}{\cos A} = 1$$

$$\tan A = 1$$

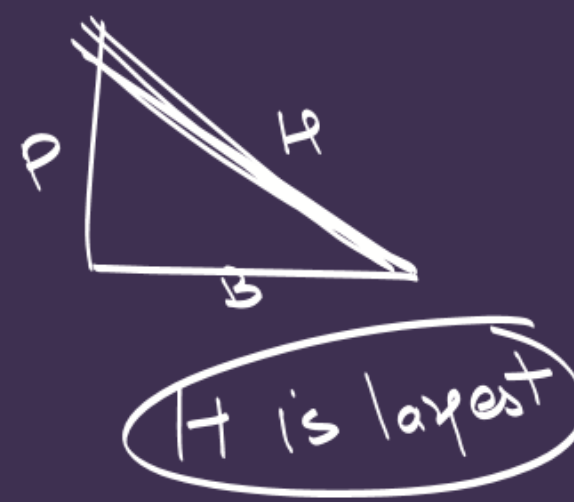
$$A = 45^\circ \checkmark$$

$$\Rightarrow \frac{(\cos 45)^\circ + (\sin 45)^\circ}{\tan 45}$$

$$\Rightarrow \frac{\left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2}{1}$$

$$\Rightarrow \frac{\frac{1}{2} + \frac{1}{2}}{1} \Rightarrow \frac{1}{1} \Rightarrow \textcircled{1}$$

#LP: State whether the following are true or false.



(i) The value of $\tan A$ is always less than 1. (F)

(ii) $\sec A = 12/5$ for some value of angle A. (T)

$$\sec A = \frac{H}{B}$$

$H > B$

$\sec A = \frac{12}{5}$ here $H > B$ ✓ correct

(iii) $\cos A$ is the abbreviation used for the cosecant of angle A. (F)

(iv) $\cot A$ is the product of cot and A. (F)

(v) $\sin \theta = 4/3$ for some angle θ . (F)

$$\frac{P}{H}$$

$$\frac{H}{P}$$

$$\frac{P}{H}$$

since H is smaller.

$\therefore$ Not possible

$\sin = \text{sine}$
 $\cos = \text{cosine}$
 $\tan = \text{tangent}$
 $\operatorname{cosec} = \text{cosecant}$
 $\sec = \text{secant}$
 $\cot = \text{cotangent}$

ATK² :-

$\begin{matrix} \sin & \cos & \tan \\ P & B & P \\ H & H & B \\ \csc & \sec & \cot \end{matrix}$

$$\sin \theta = \frac{1}{\csc \theta}$$

$$\cos \theta = \frac{1}{\sec \theta}$$

$$\tan \theta = \frac{1}{\cot \theta}$$

reciprocal
identities

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$

	0°	30°	45°	60°	90°
✓ sin θ	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
cos θ	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
✓ tan θ	0	$\frac{1}{\sqrt{3}}$	1	$\frac{\sqrt{3}}{1}$	∞
cosec θ	∞	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1
sec θ	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	∞
cot θ	∞	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0

#LP: State whether the following are true or false.

(i) $\sin(A + B) = \sin A + \sin B$. (F)

(ii) The value of $\sin \theta$ increases as θ increases. (T)

(iii) The value of $\cos \theta$ increases as θ increases. (F)

(iv) $\sin \theta = \cos \theta$ for all values of θ . (F)

(v) $\cot A$ is not defined for $A = 0^\circ$. (T)

THANK YOU COODIESS !!

